

SHONT'S AGCM, AB, Canada

WMO: 714050

Lat: 53.3325N

Lon: 112.5395W

Elev: 2228

StdP: 13.55

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-25.9	-20.2	-32.0	1.0	-25.9	-25.9	1.4	-20.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.9	83.2	62.9	80.1	62.1	77.1	61.1	67.1	77.9	64.8	75.5	62.8	73.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	94.1	73.0	60.7	86.2	69.7	58.7	80.0	67.6	32.9	77.9	31.0	75.7	29.5	73.4	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-33.7	89.3	6.2	3.0	-38.2	91.4	-41.8	93.2	-45.3	94.9	-49.8	97.2
			WB	-33.9	71.7	6.2	2.3	-38.4	73.4	-42.0	74.7	-45.5	76.0	-50.0	77.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	12.7	10.8	22.3	37.0	50.4	58.0	61.7	59.9	51.1	38.6	22.8	10.3		
	DBStd	21.91	15.57	14.91	14.10	9.50	7.85	5.34	4.70	5.64	7.92	8.62	12.81	14.87		
	HDD50	6104	1157	1097	858	399	93	7	0	2	80	364	816	1231		
	HDD65	10469	1622	1517	1323	841	454	218	123	173	417	819	1266	1696		
	CDD50	1154	0	0	0	8	105	246	362	309	114	10	0	0		
	CDD65	45	0	0	0	0	1	7	20	15	2	0	0	0		
	CDH74	1400	0	0	0	5	120	187	441	435	208	4	0	0		
	CDH80	287	0	0	0	0	20	27	92	95	53	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	16.3	0.7	0.4	0.7	1.2	1.7	2.8	3.2	2.1	1.3	0.7	0.7	0.6		
(15) Precipitation	PrecMax	22.6	2.7	0.9	1.5	2.6	4.6	5.0	5.5	4.1	3.2	1.7	3.0	1.5		
	PrecMin	8.7	0.1	0.0	0.2	0.1	0.3	0.8	1.2	0.4	0.2	0.0	0.0	0.1		
	PrecStd	2.9	0.6	0.2	0.3	0.6	0.9	1.1	1.1	0.7	0.8	0.4	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	43.2	54.4	72.3	83.0	83.3	87.3	87.5	86.6	72.6	56.0	42.8		
		MCWB	37.3	37.5	43.5	51.1	56.8	62.4	66.2	64.0	61.5	53.9	45.1	36.1		
	2%	DB	40.6	39.1	48.0	65.3	77.2	79.1	82.3	82.4	79.9	63.6	47.9	38.8		
		MCWB	35.5	34.3	39.9	47.9	54.8	60.7	65.8	62.5	58.7	49.8	40.1	33.9		
	5%	DB	37.8	35.9	43.0	58.9	72.7	75.4	78.9	78.9	74.6	58.2	42.8	34.9		
		MCWB	33.5	32.5	37.5	44.6	53.1	59.1	64.5	63.0	57.0	46.4	36.9	30.9		
	10%	DB	34.6	32.8	39.1	53.7	68.2	71.9	75.5	75.5	68.5	53.3	39.0	30.9		
		MCWB	31.4	30.1	34.3	42.0	50.9	57.2	62.7	61.9	54.4	44.1	34.9	27.9		
	0.4%	WB	38.2	37.8	44.9	52.0	58.5	66.2	71.0	69.8	62.6	55.0	46.1	36.7		
		MCDB	43.5	42.8	52.5	71.6	75.8	77.9	81.5	77.8	78.2	69.7	55.5	41.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.9	34.9	40.7	48.3	56.2	63.1	67.9	66.8	60.0	50.9	41.0	34.0		
		MCDB	40.1	38.5	47.4	64.1	72.8	74.1	78.7	77.4	77.5	61.0	46.7	38.5		
	5%	WB	34.0	32.7	37.4	45.4	54.4	60.9	65.6	64.4	57.6	47.7	37.7	31.2		
		MCDB	37.6	35.6	43.1	57.4	69.8	71.2	76.1	75.4	72.3	56.5	42.1	34.8		
	10%	WB	31.4	30.1	34.7	42.6	52.4	59.0	63.5	62.3	55.4	44.9	35.1	27.9		
		MCDB	34.7	32.9	38.6	52.0	65.5	69.5	73.4	73.1	66.9	52.5	38.5	30.8		
	5% DB	MDBR	18.6	19.4	18.8	21.7	25.9	23.2	23.9	25.8	26.2	22.1	17.9	17.6		
		MCDBR	19.0	20.1	21.8	32.7	33.5	29.4	30.9	32.8	38.5	30.8	22.0	20.9		
		MCWBR	15.0	16.3	15.0	17.3	15.0	13.8	16.1	16.6	18.9	18.1	15.5	17.0		
		MCWBR	19.2	19.6	21.1	30.2	30.2	25.5	28.1	28.9	35.2	28.9	21.2	20.5		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.259	0.307	0.331	0.347	0.361	0.359	0.369	0.311	0.279	0.253	0.246		
		taud	2.303	2.369	2.326	2.363	2.372	2.357	2.395	2.333	2.478	2.541	2.443	2.276		
	Ebn at Noon		233	275	279	286	286	282	280	270	277	266	239	213		
		Edh at Noon	18	23	31	34	36	37	35	35	27	20	16	16		
(44) All-Sky Solar Radiation	RadAvg		318	649	1133	1485	1821	1917	1971	1621	1119	649	314	234		
	RadStd		35	51	86	142	129	123	96	112	116	75	35	22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (83 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page